Pranav Vivek Malpure

858-247-8076 | pranavmalpure2020@gmail.com | github.com/pranav-malpure | pranav-malpure.github.io

EDUCATION

- **University of California San Diego** *(Sept'24 - Mar'26)*
  Master of Science in Electrical and Computer Engineering | Intelligent Systems, Robotics & Controls *GPA:* ***3.64/4***
  Courses: Statistical Learning-I, Introduction to Robotics, Linear Systems Theory, Sensing/Estimation in Robotics, Linear Algebra, Visual Learning, Planning/Learning in Robotics
- **Indian Institute of Technology Bombay, India** *(Nov'20 - Aug'24)*
  Bachelor of Technology with Honors, Aerospace Engineering *GPA:* ***8.06/10***
  Minor in Systems & Controls Engineering
  Courses: Navigation & Guidance of UAVs, Embedded Robotics, Reinforcement Learning, Intelligent Feedback & Control
  ***Achievements**: Ranked **1981** in India out of **250,000** candidates in the Joint Entrance Examination(Advnaced) (2020)*

WORK EXPERIENCE

Robotics Software Engineer | *Labelbox* | *San Francisco, CA* *(Jul'25 - Present)*

- Designed and implemented a modular **robot-learning framework** from scratch supporting interchangeable policies, robots, controllers, dataloaders, and logging systems for rapid experimentation across manipulation research workflows
- Trained and deployed $\pi 0$ with **LoRA** on in-house bimanual teleoperated demonstration data; achieving successful bowl grasping, picking and stacking behavior in **live deployment** on physical xarm6 robots
- Fine-tuned and trained **R3M** vision encoder from **scratch** on proprietary egocentric datasets; achieved **25.3%** task success on Relocate task in Adroit sim vs. **16%** baseline with open-source data, validating dataset quality
- Developed complete end-to-end **VR** based **teleoperation** data collection pipeline for bimanual robotic arms using **ROS2**
- Built end-to-end egocentric data post-processing pipeline; converting diverse capture formats (wrist, overhead stereo) into MCAP, adding action labels, and QAing pilot deliverables for customers

Robotics Intern | *Flytbase Labs* | *Pune, India* *(Jun'23 - Jul'23)*

- **Optimized** real-time addition of NFZs resulting in **reduction** of computing time by **92%** by grouping visibility graphs
- Developed an algorithm that assesses reachability of subsequent waypoints online and optimizes return-to-home decisions

KEY PROJECTS

RL for Robotic Manipulation | *Existential Robotics Laboratory, UC San Diego* *(Oct'24 - Present)*
Graduate Student Researcher

- Integrated **DrQ-v2**'s image-based data augmentation techniques into the **SAC** policy for a PickCube task in **ManiSkill**
- Implemented RL policy for the **16** joint Allegro hand to enable it to pick a cube by tuning rewards in a staged manner
- Deployed **demonstration-augmented** reinforcement learning policies on a **real** xarm6 with allegro hand, leveraging hybrid offline-to-online RL strategies for sample-efficient learning; actively improving stability and robustness
- Worked on implementing 3D **diffusion** policy for combining 3D data and denoising actions trained on imitation learning

Perception based Pedestrian Intent Prediction | *UC San Diego* *(Apr'25 - Present)*

- Developed a pedestrian intent prediction model achieving up to **88%** F1 score utilizing VGG-16 for feature extraction and a Convolutional LSTM for spatio-temporal dynamics
- Boosted prediction accuracy by using a **learning rate scheduler** & experimenting with different input sequence lengths
- Enhanced temporal analysis by integrating pedestrian bounding box and YOLO-Pose derived body pose data into a novel LSTM-based architecture for binary intent classification

The Humanoid Project | *Student Tech Team, IIT Bombay* *(Mar'22 - Apr'24)*
Team Lead

- Led a team of **20** students building a full sized humanoid robot to be deployed for **sorting** books in the central library
- Crafted roadmaps to ensure technical coordination between subsystems & oversaw budget allocation of INR **0.2 million**

Autonomous Navigation of UUVs | *Aerospace Dept., IIT Bombay* *(Jan'23 - Apr'23)*

- Implemented the **curvature velocity method** in python to navigate a UUV through static obstacles using ROS-Gazebo
- Leveraged data from **3** onboard **sonar** sensors to detect obstacles, enabling **real-time** adjustments of thrust & velocity

TECHNICAL SKILLS

Languages/Frameworks C++, Python, MATLAB, Robot Operating System (ROS), ROS 2, Git, Embedded Linux
Packages and Libraries Numpy, Pandas, SciPy, NLTK, pyvisgraph, Pytorch
Softwares and Simulators Gazebo, RViz, dm_control-MuJoCo, Maniskill-SAPIEN